Module 3 -- Integrated Circuits (ICs)

Lesson 23: Building a Simple Counter

★ Star-3 Intermediate ⏰ 45--50 minutes 👤 Age 6--12 years ⚡ 300 XP

Your Mission Today

Get ready for one of the coolest circuits in this module! Today you will combine TWO ICs — the 555 timer you already know and a brand new chip called the **4017 decade counter**. The 555 generates pulses, and the 4017 counts them. The result? Ten LEDs that light up one at a time in sequence, like a light show! You are about to build your own LED chaser.

Learning Objectives

By the end of this lesson, you will be able to:

- Explain what a counter IC does (counts pulses and activates outputs in sequence)
- Identify the 4017 decade counter IC pins
- Build a 555 + 4017 LED chaser circuit
- Control the chaser speed by adjusting the 555 timing components
- Use your Magic Measurement Wand to trace which output is active

What You Need

Item	Qty
555 timer IC	1
4017 decade counter IC	1
LEDs (any colors — a mix is fun!)	10
330-ohm resistors	10
Resistor: 1k-ohm (R1 for 555)	1
Resistor: 10k-ohm (R2 for 555)	1
Resistor: 47k-ohm (R2 swap for slower speed)	1
Electrolytic capacitor: 10 microfarad (C1 for 555)	1
Electrolytic capacitor: 47 microfarad (C1 swap)	1
Ceramic capacitor: 0.01 microfarad (noise filter)	1
9V battery + clip	1
Breadboard (830-point recommended)	1
Jumper wires	25
Multimeter (Magic Measurement Wand!)	1

How to Teach This Lesson

Step 1: Hook — The Knight Rider Light Show (5 min)

"Have you ever seen those movie cars with lights that sweep back and forth across the front? Or a row of lights on a stage that chase each other? That is called a light chaser, and it is exactly what we are building today!"

"But here is the clever part — we are not going to use a computer. We are using just TWO chips: the 555 timer to generate a clock signal (tick... tick... tick...) and the 4017 counter to count each tick and light up the next LED."

Draw a simple picture:

```
555 Timer        4017 Counter         LEDs
[tick tick] ---> [0, 1, 2, 3...] ---> [LED0] [LED1] [LED2] ...
(clock)          (counts pulses)       (light up in sequence)
```

Award: **+10 XP** **for understanding how two ICs work together!**

Step 2: Meet the 4017 Decade Counter (8 min)

What is a decade counter?

"A decade counter is a chip that counts from 0 to 9 (that is 10 steps — a decade means ten). Every time it receives a pulse on its CLOCK pin, it moves to the next output. Output 0 goes HIGH, then output 1, then output 2, and so on. When it reaches output 9, it loops back to 0 and starts over."

Key vocabulary:

- **Clock signal:** A repeating pulse (HIGH-LOW-HIGH-LOW) that tells the counter to advance one step

- **Carry out:** A signal that goes HIGH after the counter completes a full cycle (useful for chaining counters)

The 4017 IC pinout:

```
4017 Decade Counter -- Top View

      Notch
        U
  +----------+
  |  1    16 | <-- Pin 16 = VCC (+9V)
  |  2    15 |
  |  3    14 | <-- Pin 14 = Clock input
  |  4    13 | <-- Pin 13 = Clock Enable (connect to GND)
  |  5    12 |
  |  6    11 |
  |  7    10 |
  |  8     9 |
  +----------+
     ^
  Pin 8 = GND

Output pins:
Pin 3  = Output 0 (Q0)
Pin 2  = Output 1 (Q1)
Pin 4  = Output 2 (Q2)
Pin 7  = Output 3 (Q3)
Pin 10 = Output 4 (Q4)
Pin 1  = Output 5 (Q5)
Pin 5  = Output 6 (Q6)
Pin 6  = Output 7 (Q7)
Pin 9  = Output 8 (Q8)
Pin 11 = Output 9 (Q9)

Pin 15 = Reset (connect to GND for normal counting)
Pin 12 = Carry Out
```

> "Notice the pin numbers are not in order — they jump around! This is normal for ICs. Always check the datasheet."

Award: **+10 XP** **for learning the 4017 pin layout!**

Step 3: Build the 555 Clock Circuit First (8 min)

"Before we wire the counter, let us build the 555 timer in astable mode. This will be our clock — the heartbeat that tells the counter when to step forward."

555 Astable Circuit (same as Lesson 18, but faster):

```
+9V ----+
        |
       [R1] 1k-ohm
        |
        +---- Pin 7 (Discharge)
        |
       [R2] 10k-ohm
        |
        +---- Pin 6 (Threshold) + Pin 2 (Trigger)
        |
       [C1] 10uF
        |
       GND

Pin 8 = VCC (+9V)
Pin 1 = GND
Pin 4 = +9V (Reset -- keep enabled)
Pin 5 = 0.01uF to GND (noise filter)
Pin 3 = Output (goes to 4017 clock input)
```

With R1=1k, R2=10k, C1=10uF:

```
Frequency = 1.44 / ((1000 + 20000) x 0.00001)
          = 1.44 / 0.21
          = about 6.9 Hz (roughly 7 blinks per second)
```

"That is fast enough to see the LEDs chase but slow enough to follow with your eyes."

Test: Connect an LED to Pin 3 temporarily to confirm the 555 is working before connecting the 4017.

Award: **+20 XP** **for building the working 555 clock!**

Step 4: Wire the 4017 Counter and LEDs (12 min)

"Now let us add the counter chip and all 10 LEDs. This is the biggest circuit we have built so far! Take it step by step."

Step-by-step wiring:

1. Place the 4017 IC on the breadboard (on the same board or a second one connected with jumper wires)
2. Pin 16 (VCC) to + rail
3. Pin 8 (GND) to - rail
4. Pin 13 (Clock Enable) to - rail (GND = counting enabled)
5. Pin 15 (Reset) to - rail (GND = no reset, count normally)
6. Pin 14 (Clock Input) to **Pin 3 of the 555** (the clock signal!)

Now wire the 10 LEDs:

For each output pin on the 4017, connect:

- Output pin —> 330-ohm resistor —> LED (long leg +) —> LED (short leg) —> GND rail

```
| 4017 Pin | Output | LED Color (your choice) |
|----------|--------|-------------------------|
| Pin 3    | Q0     |                         |
| Pin 2    | Q1     |                         |
| Pin 4    | Q2     |                         |
| Pin 7    | Q3     |                         |
| Pin 10   | Q4     |                         |
| Pin 1    | Q5     |                         |
| Pin 5    | Q6     |                         |
| Pin 6    | Q7     |                         |
| Pin 9    | Q8     |                         |
| Pin 11   | Q9     |                         |
```

"Connect the battery. Watch the LEDs! They should light up one at a time, chasing across the row. LED 0 lights, then LED 1, then LED 2... all the way to LED 9, then it starts over!"

Award: **+40 XP** **for building the full 10-LED chaser circuit!**

Step 5: Speed Control (5 min)

"Let us play DJ again! Change the 555 components to adjust the chase speed."

Experiment:

```
| R2 Value | C1 Value | Expected Speed | Looks Like |
|----------|----------|---------------|------------|
| 10k      | 10uF     | Fast (~7 Hz)  |            |
| 47k      | 10uF     | Medium (~1.5 Hz) |         |
| 10k      | 47uF     | Slow (~1.5 Hz) |           |
| 47k      | 47uF     | Very slow (~0.3 Hz) |      |
```

"Make it go as fast as you can! Then slow it down so you can count each step. Which speed looks coolest?"

Award: **+20 XP** **for testing different speeds!**

Step 6: Wand Check — Trace the Active Output! (8 min)

"Only one output is HIGH at any time. Let us prove it with the Wand!"

Slow the circuit down (use R2=47k, C1=47uF for very slow stepping).

1. Set the Wand to DC Voltage
2. Black probe on GND
3. Touch the red probe to each 4017 output pin, one at a time

What you should see:

- Most pins read close to **0V** (LOW — LED off)
- One pin reads close to **9V** (HIGH — LED on)

> 💬 "At any moment, ONLY ONE output is HIGH. The counter moves the HIGH signal from one pin to the next with each clock pulse. It is like passing a baton in a relay race!"

Activity — Find the active pin:

```
| Clock Tick | Which Pin is HIGH? | Wand Voltage |
|------------|--------------------|--------------|
| Tick 1     |                    |              |
| Tick 2     |                    |              |
| Tick 3     |                    |              |
| Tick 4     |                    |              |
| Tick 5     |                    |              |
```

Bonus: Measure the clock signal

Touch the red probe to Pin 14 (Clock Input). Watch the voltage flip between HIGH and LOW with each 555 pulse.

Award: **+50 XP** **for tracing the active output across 5 ticks!**

Fun Game: Light Show Patterns (Optional Extension)

Instead of 10 separate-color LEDs, try patterns:

- Alternating red and green for a holiday theme
- All one color for a smooth wave effect
- Use only 5 outputs (connect Pin 5/Q6 Reset to stop at 5 and loop)

To make a 5-LED loop: Connect the output of Q5 (Pin 1) to the Reset pin (Pin 15). Now the counter resets after 5 instead of 10!

Quick Quiz — Earn Bonus XP!

Question 1: What does the 4017 decade counter do?

- B) Counts clock pulses and activates one output at a time, 0 through 9
- C) Measures voltage

(Correct: B — +20 XP !)

Question 2: What provides the clock signal to the 4017 in our circuit?

- A) The battery
- B) The 555 timer in astable mode
- C) The LEDs

(Correct: B — +20 XP !)

Question 3: At any given moment, how many 4017 outputs are HIGH?

- A) All 10
- B) 5
- C) Just 1

(Correct: C — +20 XP !)

Question 4: To make the LED chase SLOWER, you should:

- A) Add more LEDs
- B) Use a bigger R2 or bigger C1 in the 555 circuit
- C) Remove the 4017

(Correct: B — +20 XP !)

Lesson 23 Complete!

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   LIGHT CHASER CHAMPION BADGE UNLOCKED!

   Skills unlocked:
   [check] Understand the 4017 decade counter
   [check] Combined 555 timer + 4017 counter
   [check] Built a 10-LED chaser circuit
   [check] Controlled chaser speed
   [check] Traced active outputs with the Wand

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```

XP Breakdown:

Activity	XP
Two-IC concept understanding	10
Learn 4017 pinout	10
Build 555 clock circuit	20
Build full 10-LED chaser	40
Speed control experiments	20
Wand Check (trace 5 ticks)	50
Quiz (4 questions)	80
TOTAL POSSIBLE	**230**

Coming Up Next...

In **Lesson 24** — the **Module 3 Grand Finale** — you will build a **Digital Dice**! Press a button and a random number from 1 to 6 appears. It uses everything you have learned: the 555 timer, the 4017 counter, and LEDs arranged in a dice pattern. Let us roll!

Troubleshooting

Problem	Fix
LEDs do not chase — all off	Check that Pin 14 (Clock) is receiving the 555 output
LEDs do not chase — one stays on	Pin 13 (Clock Enable) must be connected to GND
Counter seems stuck at 0	Pin 15 (Reset) must be connected to GND for normal counting
LEDs chase but skip some	Check the wiring from each output pin — the pin numbers are NOT in sequential order
Chase is too fast to see	Use bigger R2 and/or bigger C1 in the 555 circuit
Only some LEDs light up	Check each LED polarity and that each has its own 330-ohm resistor
One LED lights but is dim	That LED might be inserted backwards — check polarity (long leg = +)